

GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

QIAN QI

ABSTRACT. A finite causal observer must choose experiments as well as compress their reports. We prove a completion theorem for observation-adaptive experiments with overlapping noise. The construction pools posterior laws by their actual conditional barycenters, so every retained state has its true posterior rather than an uncharged exact filter or an approximate posterior. Its excess decision risk is exactly a sum of acquired Jensen defects of concave continuation values. An integrated curvature estimate makes these defects second order under uniform posterior-density bounds, without differentiability of the optimal value or a margin assumption at policy switches. This yields matching checkpoint and autonomous memory exponents at every fixed observation horizon, with the complete phase and controller cost displayed. Positive finite-state kernels with continuous asymmetric sensors verify the assumptions, and the optimal sensor changes with a noisy observed value. A stratified version permits exact-validation atoms and continuously acquired strata in the same recursion. Calibration, numerical implementation and causal deficiency are transferred in the same task. The separated-refinement and continuation-cut theorems are retained in integral supplements; uniform joint horizon-memory optimality outside those stronger classes is not asserted.

1. THE CAUSAL COMPLETION PROBLEM

A preparation and its legal experimental kernels determine which future tests can actually be performed. They do not supply a manifold, a Fisher matrix, or an observer with an exact posterior register. The question is to connect the predictive information obtained from these tests to a causal implementation whose acquisition controller and memory are both charged. We use *General Theta* for this mathematical programme: prepared kernels, executable tests, a measurable predictive quotient, acquired geometry, certified causal comparison, and decision risk. Generality means a theorem on a stated class, not that every experiment possesses every regularity property.

There are two different obstacles. An arbitrary small terminal code may discard information needed by a later experiment; continuation responses reveal that obstruction. Even when a small code can preserve the relevant response, the observer must decide its next experiment without consulting the discarded history. Strongly separated response refinements resolve the second obstacle by tree pruning, but ordinary noisy reports have overlapping conditional supports. An informative report is not automatically a separated refinement.

Theorem 2.2 treats the latter case without a support gap. It constructs a causal finite-state observer whose retained labels carry their *actual* posterior distributions. After an action and a report, all incoming posterior values assigned to a cell are pooled together. The cell's read-only representative is their conditional mean under the law induced by this very observer. A Bellman identity then expresses the loss of information exactly as acquired Jensen defects. Concavity controls those defects even where the optimal action switches discontinuously. Lemma 2.3 proves the required integrated second-order estimate for nonsmooth concave values.

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Write B_n^* for the optimal full-history risk with n observations. If every one-step posterior law, uniformly over incoming beliefs and legal actions, has bounded density on a compact interior d -dimensional set, then

$$(1) \quad R_{\text{cp}}(n, M) - B_n^* \asymp M^{-2/d}, \quad R_{\text{aut}}(n, M) - B_n^* \asymp_n M^{-2/d}.$$

The first constants are uniform in the horizon when the kernel bounds are. For the second statement the quantitative upper is

$$Cn \left\lfloor \frac{M-1}{n} \right\rfloor^{-2/d};$$

the machine has at most $1 + n \lfloor (M-1)/n \rfloor \leq M$ states, including every phase. Thus (1) is a fixed-horizon completion law, not a uniform joint n, M equivalence. The full-history acquisition policy is optimized, rather than fixed and subsequently compared with a different control problem.

Theorem 3.1 verifies the posterior geometry from positive finite-state transitions and continuous report densities by an explicit change of variables. In a two-state specialization, the best last sensor switches at posterior probability 1/2. A two-call feedback policy has strictly lower risk than every fixed two-sensor sequence. No finite report excludes either hidden state. Theorem 4.1 allows positive-dimensional acquired laws and atoms in one controller, and its validation-sensor realization repeatedly changes between those ranks. These are consequences of the general pooling theorem, not its premises.

The complete separated-acquisition and continuation programmes are supplied as integral Supplement T [13]; the earlier block-stability, stopped-task and instrument proofs remain in integral Supplement S [14]. No substantive proof has been deleted. Supplement T gives a horizon-uniform completion law on separated response trees, together with distinct noisy continuation obstructions; the present paper closes the overlapping-noise adaptive case on a new class, with the horizon dependence above. We do not claim an intrinsic necessary-and-sufficient classification of all causal experiments, an optimal planning algorithm, or a matched region for every computational resource.

1.1. Experiments, tests and the state model. All spaces are standard Borel. A prepared experiment has probability kernels

$$K_t^\lambda(dx', dy, dc \mid x, a),$$

where c records physical calls and time. Policies and simulators are independent of the unknown parameter λ ; all preparation laws, calibration data and policy classes used in a Bayes assertion are specified. A visible history includes actions, reports, publicly supplied costs and legal interface information. An executable future test is a legal finite continuation followed by a bounded scored readout. Histories are predictively equivalent when their legal interfaces and every such expectation agree. Appendix A records a measurable realization and its exact minimality condition. A task realization need not simulate the complete experiment.

At a persistent cut an autonomous machine has only its current label in a fixed set of at most M states. Its next action, stop instruction and final readout depend on that label alone; its transition may also use the current report and fresh independent randomness. No elapsed time, program position, earlier output, old report, or private seed remains available unless counted in the state. Read-only instructions may depend on the known experiment and budgets, not the unknown parameter or acquired data. Temporary computation is erased at the next cut. Abstract Borel instructions and known real constants are allowed in the cardinality theorem; Section 5 separately gives a certified finite-precision implementation. Existence of the programme is not efficient synthesis of it.

A fixed n -call task permits actions at each of the n stages and reveals the scored audit only after the forecast. This theorem does not optimize the physical stopping time: its phase is internal to the finite machine. Bounded stopping with a nonintervening audit has a separate result in Supplement T; no fixed-time statement is substituted at an arbitrary stopping time here. A checkpoint procedure may use the full history during acquisition and retain only M labels at its terminal cut. Both procedure classes use the same preparation, actions and audit.

For a finite measure ν in a Euclidean task space set

$$(2) \quad q_M(\nu) = \inf_{c_1, \dots, c_M} \int \min_j \|x - c_j\|^2 \nu(dx).$$

Subprobability mass is not normalized away. For a fixed deterministic acquisition policy π , let $P_n^\pi = \mathbb{E}[Y \mid H_n^\pi]$, $B_n^\pi = \mathbb{E}\|Y - P_n^\pi\|^2$, and $\nu_n^\pi = \text{Law}(P_n^\pi)$. Projection gives the fixed-policy identity

$$(3) \quad \inf_{M\text{-label terminal codes}} \mathbb{E}\|Y - \hat{Y}\|^2 = B_n^\pi + q_M(\nu_n^\pi).$$

The encoder assigns the least-index nearest center; such ties are Borel. An infimum suffices if an optimal codebook is not attained.

Lemma 1.1 (Independent randomization at a checkpoint). *Suppose all acquisition and coding coins are independent of the prepared physical randomness. Allowing a full-history policy and its checkpoint code to use these coins does not lower the infimum in (3) after joint optimization over deterministic policies and codes. In particular,*

$$(4) \quad R_{\text{cp}}(n, M) = \inf_{\pi \text{ deterministic}} \{B_n^\pi + q_M(\nu_n^\pi)\}.$$

Proof. Supply the encoder and decoder with the entire independent seed, as a relaxation if necessary. For each fixed seed their full-history policy and code are deterministic; their risk is bounded below by the deterministic joint infimum. Integrate over the seed. Deterministic procedures are admitted in the randomized class, proving equality. This argument concerns the full-history checkpoint problem. Fixing a time-indexed coin tape need not preserve the autonomous state budget, and no such purification is asserted. For an autonomous squared-loss lower bound one may first replace each randomized terminal readout by its conditional mean, obtaining at most M fixed centers, and then relax its earlier control to full-history control. $\square$

The policy infimum in (4) is not interchanged with either of its policy-dependent summands.

2. CAUSAL POSTERIOR POOLING

Fix $d \geq 1$, a horizon $n \geq 1$, and hidden states $\{0, \dots, d\}$. The preparation p_0 and all kernels below are known. At stage $t < n$ an action a in a fixed finite alphabet $\mathcal{A}_t$ produces a transition matrix T_t^a , then a report with strictly positive density $f_{t,j}^a(y)$ given the new hidden state j , relative to a specified measure m_t^a . These are normalized Borel densities. The primitive cost of this operation is one observation and its declared physical duration. The final scored audit is $Y = e_{X_n}$, revealed only after $\hat{Y} \in \Delta_d$, with squared Euclidean loss. Here Δ_d is the probability simplex on $d + 1$ points.

For $p \in \Delta_d$ put $q = pT_t^a$ and

$$(5) \quad r_t(p, a, y) = \sum_{j=0}^d q_j f_{t,j}^a(y), \quad F_{t,j}(p, a, y) = \frac{q_j f_{t,j}^a(y)}{r_t(p, a, y)}.$$

The report law is $r_t(p, a, y)m_t^a(dy)$. The posterior law generated by this actual report is denoted $\Lambda_t(p, a)$. Densities of posterior laws below use coordinates $(p_1, \dots, p_d)$, with $p_0 = 1 - \sum_{j=1}^d p_j$, and ordinary d -dimensional Lebesgue measure in these coordinates.

Definition 2.1 (Uniform acquired smoothing). There are $0 < \kappa < 1/(d+1)$ and $\rho < \infty$, fixed before the policy, for which, for every $t < n$, $p \in \Delta_d$ and $a \in \mathcal{A}_t$, the law $\Lambda_t(p, a)$ is supported on

$$K_\kappa = \{u \in \Delta_d : u_j \geq \kappa \ (0 \leq j \leq d)\}$$

and has density at most ρ . The same requirement includes beliefs arising after earlier compression, not just a chosen exact-filter trajectory.

This is an experiment-level condition on the pushforward in (5); Theorem 3.1 derives it by a raw Jacobian calculation. No optimal codebook, known acquired law of a chosen policy, smooth value function, or separated response support is assumed. The report law itself obeys

$$(6) \quad \text{TV}(J_t(p, a), J_t(p', a)) \leq \frac{1}{2} \|(p - p')T_t^a\|_1 \leq \frac{1}{2} \|p - p'\|_1.$$

The substitute for pathwise contractivity is the exact information-loss identity below. It is decision stability on actual laws, not contraction of two filters along arbitrary reports.

Let $V_t(p)$ be optimal full-history risk for the remaining stages, with

$$(7) \quad V_n(p) = 1 - \|p\|^2, \quad V_t(p) = \min_{a \in \mathcal{A}_t} \int V_{t+1}(F_t(p, a, y)) r_t(p, a, y) m_t^a(dy).$$

Then $B_n^* = V_0(p_0)$. The action set at every stage is nonempty, and all full-history Borel policies are admitted. In particular this is not a comparison under a controller fixed before optimization.

Theorem 2.2 (Adaptive acquired-geometry completion). *Under Definition 2.1, there are positive constants c, C, L_0 depending only on d, κ, ρ with the following properties.*

For every $M \geq 1$,

$$(8) \quad cM^{-2/d} \leq R_{\text{cp}}(n, M) - B_n^* \leq CM^{-2/d}.$$

For arbitrary integers $L_1, \dots, L_n \geq L_0$, a deterministic autonomous machine exists with at most

$$(9) \quad S = 1 + \sum_{t=1}^n L_t$$

total states and risk at most

$$(10) \quad B_n^* + C \sum_{t=1}^n L_t^{-2/d}.$$

All phases, action instructions and the terminal readout are functions of these states. It never rereads a report or retains an exact full-history posterior. Its read-only programme is constructed from the known kernels and the preparation.

More precisely, let λ_t be its actual pre-pooling posterior law, partitioned into its grid cells $C_{t,j}$, and let $b_{t,j}$ be the conditional barycenter in that cell. Then its risk satisfies the exact identity

$$(11) \quad R_{\text{pool}} - B_n^* = \sum_{t=1}^n \sum_j \left\{ \lambda_t(C_{t,j}) V_t(b_{t,j}) - \int_{C_{t,j}} V_t(u) \lambda_t(du) \right\}.$$

The law λ_t is induced by the constructed controller, not borrowed from an optimal full-history policy.

Consequently, for $M \geq 1 + nL_0$, with $L = \lfloor (M-1)/n \rfloor$,

$$(12) \quad cM^{-2/d} \leq R_{\text{aut}}(n, M) - B_n^* \leq CnL^{-2/d}.$$

The lower is valid whenever the autonomous class is nonempty. Thus at every fixed horizon the checkpoint and autonomous memory exponents agree. The displayed upper, rather than a horizon-independent constant, is the joint-budget assertion.

The programme in this theorem may use exact Borel Bayes instructions and exact known barycenters. Its existence is separate from the complexity of evaluating (7). A finite-precision implementation with explicitly charged errors is given in Proposition 5.2. No polynomial planning claim is implicit.

2.1. An integrated Jensen estimate.

Lemma 2.3 (Curvature integrated against actual mass). *Let K be a compact convex subset of an open convex set $\Omega \subset \mathbb{R}^d$, with positive distance from its boundary. Let v be concave and L -Lipschitz on Ω . Partition space into half-open cubes C_j of side h , with fixed lexicographic boundary assignment. For all sufficiently small h , every finite measure λ supported on K with density at most ρ satisfies*

$$(13) \quad 0 \leq \sum_{j: \lambda(C_j) > 0} \left\{ \lambda(C_j) v(b_j) - \int_{C_j} v(x) \lambda(dx) \right\} \leq C_{K, \Omega, d} \rho L h^2, \quad b_j = \frac{\int_{C_j} x \lambda(dx)}{\lambda(C_j)}.$$

The constant does not require differentiability of v , a lower bound on cell masses, or regularity of its switching sets. The measure need not be normalized.

Proof. Jensen's inequality gives the lower. We prove the upper first for smooth convex $f = -v$. For a cube C meeting K , let z be its center, let ℓ be the tangent affine function to f at z , and set $w = f - \ell$. Take $R = 2\sqrt{d}h$; the ball $B_R(z)$ remains in a fixed compact neighborhood of K when h is sufficiently small. Convexity gives $w \geq 0$, $w(z) = 0$, and subharmonicity gives, for $x \in C$,

$$w(x) \leq 2^d \oint_{B_R(z)} w \leq 2^d A(R), \quad A(r) = \oint_{\partial B_r(z)} w.$$

For the first inequality average over $B_{R/2}(x) \subset B_R(z)$, using $w \geq 0$; for the second, the spherical averages increase. Along each ray convexity and $w(z) = 0$ give $w(z + R\theta) \leq R \partial_R w(z + R\theta)$. The divergence theorem therefore yields

$$(14) \quad \sup_C w \leq C_d R^{2-d} \int_{B_R(z)} \Delta f(x) dx.$$

Since ℓ is affine and $w(b_C) \geq 0$, the cell's Jensen gap for v is at most

$$\int_C w d\lambda \leq C_d \rho h^2 \int_{B_R(z)} \Delta f(x) dx.$$

The expanded balls of a lattice grid have bounded overlap depending only on d . Choose a smooth cutoff χ equal to one on their union and supported in a fixed compact subset of Ω . Positivity of Δf and integration by parts give

$$\int_{\cup B_R(z)} \Delta f \leq \int \chi \Delta f = - \int \nabla \chi \cdot \nabla f \leq L \int \|\nabla \chi\|.$$

This proves (13). For a nonsmooth convex f , mollify on a slightly larger compact neighborhood. The mollifications remain convex, have Lipschitz constants at most L , and converge uniformly on all cells and their barycenters. The same bound is uniform in the smoothing radius; passage to the limit proves the assertion. Equivalently, the estimate integrates the positive distributional curvature measure of f , rather than bounding its pointwise Hessian. $\square$

Here and below the notation $\oint$ denotes normalized integration over the indicated set. The argument is a local convexity estimate; no claim of novelty is made for concavity, subharmonic mean inequalities or mollification individually. Its role is to control causal pooling for nonsmooth optimal continuation values.

2.2. Exact posterior states and optimal actions.

Lemma 2.4 (Finite-state posterior aggregation). *Suppose a finite label I has known masses α_i and exact conditional hidden distributions $p_i = \text{Law}(X \mid I = i)$. Select a known action a_i at each label, take the physical transition and report, and compute $W = F(p_i, a_i, Y)$. If the next label is $J = j$ precisely when $W \in C_j$, then*

$$(15) \quad \alpha'_j = \mathbb{P}(J = j) = \lambda(C_j), \quad \text{Law}(X' \mid J = j) = b_j = \frac{\int_{C_j} u \lambda(du)}{\lambda(C_j)}, \quad \lambda = \text{Law}(W).$$

Thus the barycenter pools all incoming labels, not separate incoming edges. Zero-mass cells may be assigned arbitrary fixed instructions. The next transition uses only J , its recorded barycenter and the new report.

Proof. The Markov kernel and $p_i = \text{Law}(X \mid I = i)$ imply $\text{Law}(X' \mid I, Y) = W$. For each hidden coordinate k ,

$$\mathbb{P}(X' = k, J = j) = \mathbb{E}[W_k \mathbf{1}_{\{W \in C_j\}}] = \int_{C_j} u_k \lambda(du).$$

Divide by the cell's actual positive mass. This proves the exact conditional identity without retaining I or its history at the next cut. $\square$

Proof of Theorem 2.2. A fixed future policy, including its final simplex-valued forecast, has risk linear in the incoming hidden distribution, with coefficients in $[0, 2]$. Taking the infimum shows that every V_t is concave and $2\sqrt{d}$ -Lipschitz in simplex coordinates. The Bayes recursion (7) follows by conditioning. Its integrands are bounded, and the perspective of a continuous bounded function under (5) is continuous in p by dominated convergence; an integrable dominator is $2 \sum_j f_{t,j}^a$. Finite action sets therefore admit Borel minimizing selectors, with ties resolved by a fixed order.

Start with one initial label of mass one and posterior p_0 . Inductively, at a label with posterior p_i , choose a minimizing action in (7). Partition the next posterior's first d coordinates into the grid of side $h_t = 1/k_t$, where $k_t = \lfloor L_t^{1/d} \rfloor$. There are at most $k_t^d \leq L_t$ occupied cells in $[0, 1]^d$. Apply Lemma 2.4 to all incoming labels and record the resulting masses and barycenters in the read-only programme. This is a finite forward construction of integrals under the actual controller law. It is not an oracle that supplies the true full-history posterior during execution.

Let Z_t denote the retained label and $Q_t = \text{Law}(X_t \mid Z_t)$ its recorded posterior. Let W_t be the pre-pooling posterior using (Z_{t-1}, Y_t) . Bellman optimality at the chosen action gives

$$\mathbb{E}V_t(W_t) = \mathbb{E}V_{t-1}(Q_{t-1}).$$

Equation (15) gives $Q_t = \mathbb{E}[W_t \mid Z_t]$. Summing the successive differences proves (11), because the terminal output Q_n has true risk $\mathbb{E}[1 - \|Q_n\|^2]$. In general W_t is *not* the posterior given the entire discarded history. No identification with that posterior is used.

By Definition 2.1, each λ_t is a finite mixture of laws bounded by ρ on K_κ . Choose L_0 so that the grids in Lemma 2.3 and their enlarged cubes stay in the simplex interior. That lemma bounds the t th summand by $CL_t^{-2/d}$ and proves (10). Give all stages disjoint label sets, plus the initial label. Their union has size (9); the stage, action, transition routine and stop/readout instruction are determined by the label itself. This proves the full autonomous state count.

For a converse, under *any* full-history policy, condition immediately before the last action and report. Definition 2.1 bounds the resulting posterior law by ρ ; averaging gives the same bound for ν_n^π . In the first d coordinates, a union of M radius- r balls has mass at most $\rho v_d M r^d$, where v_d is the unit-ball volume. Set $r = (2\rho v_d M)^{-1/d}$. At least half the mass lies outside, giving

$$(16) \quad q_M(\nu_n^\pi) \geq \frac{1}{2} (2\rho v_d M)^{-2/d}.$$

The full simplex norm dominates the coordinate norm. Together with $B_n^\pi \geq B_n^*$ and Lemma 1.1, this proves the checkpoint lower and hence the autonomous lower. For the checkpoint upper select an optimal full-history policy and quantize its terminal posterior by a uniform grid on the simplex. There are at most M cells after a fixed dimensional adjustment, and squared cell diameter is at most $CM^{-2/d}$. Increasing C handles the finitely many small M . This proves (8). Equal allocation in (9) proves (12). $\square$

The lower bound is a bound on an actually obtained posterior law for every admissible policy. It is not a packing of a merely reachable parameter set. The upper uses global cells but charges only their finite labels and programmed constants. Its decision stability holds across arbitrary policy switches because the representatives are exact conditional barycenters. A nearest grid point maintained alongside a hidden exact posterior would neither have identity (11) nor satisfy the state model.

3. CONTROLLED SENSORS WITH OVERLAPPING NOISE

We now verify the preceding theorem from kernels rather than from prescribed posterior geometry. Let $m = d + 1$, and let every transition matrix have entries at least $\tau > 0$. Reports take values in $[-1, 1]^d$. For a reference sensor, relative to Lebesgue measure, set

$$(17) \quad f_0(y) = 2^{-d}, \quad f_j(y) = 2^{-d}(1 + \lambda y_j), \quad 1 \leq j \leq d, \quad 0 < \lambda < 1.$$

Actions may permute the hidden labels and report coordinates and may choose different positive transition matrices. A fixed finite family is selected in advance. More generally λ may range in a fixed compact subinterval of $(0, 1)$, with the constants below replaced by their uniform bounds. Each action consumes one report and one physical transition. The hidden state is not reset to a publicly revealed value.

Theorem 3.1 (Raw posterior geometry and adaptive sensor completion). *For the kernels (17), Definition 2.1 holds with*

$$(18) \quad \kappa = \frac{\tau(1 - \lambda)}{1 + \lambda}, \quad \rho = \frac{(1 + \lambda)^{d+2}}{2^d \lambda^d \tau^{d+1}},$$

or with any smaller κ . Thus Theorem 2.2 holds for every horizon and every full-history action policy in the declared family. The acquired posterior law has full dimension d after a single report; this rank is derived from the raw kernel.

If at each phase some legal transition is invertible, the hidden posterior together with that phase is a minimal predictive realization for the executable next-report tests and terminal audit. In particular there is no unverified latent parameter geometry in the dimension assertion.

For $d = 1$ take

$$T = \begin{pmatrix} 1 - \alpha & \alpha \\ \alpha & 1 - \alpha \end{pmatrix}, \quad 0 < \alpha < 1/2,$$

and two sensors $+$ and $-$, where $+$ has $(f_0, f_1) = (1, 1 + \lambda y)/2$ and $-$ has $(f_0, f_1) = (1 + \lambda y, 1)/2$. From the symmetric preparation, the optimal two-call policy changes its second sensor with the first noisy report and has strictly smaller risk than every fixed two-sensor sequence. All posterior supports overlap on both hidden states at every finite time. For each fixed n its optimal checkpoint and total-state autonomous excess risks are both of order M^{-2} .

Proof. Write $q = pT$, so $q_i \geq \tau$, and $D = 1 + \lambda \sum_{j=1}^d q_j y_j$. Then $1 - \lambda \leq D \leq 1 + \lambda$, the report density is $D/2^d$, and

$$(19) \quad u_0 = \frac{q_0}{D}, \quad u_j = \frac{q_j(1 + \lambda y_j)}{D}.$$

Every coordinate is at least κ . The map from y to $(u_1, \dots, u_d)$ has the explicit inverse

$$y_j = \lambda^{-1} \left(\frac{q_0 u_j}{q_j u_0} - 1 \right).$$

The matrix determinant lemma, or direct differentiation, gives

$$(20) \quad \det \frac{\partial(u_1, \dots, u_d)}{\partial(y_1, \dots, y_d)} = \frac{\lambda^d \prod_{j=0}^d q_j}{D^{d+1}} > 0.$$

Consequently the pushed-forward density is

$$\frac{D^{d+2}}{2^d \lambda^d \prod_{j=0}^d q_j} \leq \rho.$$

The boundary of the report cube has zero mass. This is a full-dimensional actual probability law, not a chosen volume measure on a reachable set. Permutations only change coordinates and preserve the uniform constants.

For minimality, under a reference sensor the bounded next-report tests satisfy

$$\mathbb{E}[y_j \mid p, a] = \lambda(pT^a)_j/3 \quad (1 \leq j \leq d).$$

When T^a is invertible these expectations identify p ; the terminal audit identifies its own posterior directly. Equal posteriors conversely determine all future laws by the finite-state Markov property. Finite-horizon history spaces are finite unions of compact report cubes and their posterior maps are continuous, so their images are compact Borel spaces. The interface phase is included; histories with different available continuations are not identified.

It remains to prove genuine feedback. With one call left, let q be the predicted probability of hidden state one before its report. For the plus sensor, elementary conditional variance gives the two-coordinate Brier risk

$$(21) \quad b_+(q) = 2q(1-q) - 2q^2(1-q)^2 I(q), \quad I(q) = \frac{\lambda^2}{2} \int_{-1}^1 \frac{y^2}{1 + q\lambda y} dy = \sum_{k=0}^{\infty} \frac{\lambda^{2k+2} q^{2k}}{2k+3}.$$

The uniformly convergent series follows from $\lambda < 1$. The minus sensor has $b_-(q) = b_+(1-q)$. All nonconstant coefficients of I are positive, whence

$$(22) \quad b_-(q) - b_+(q) = 2q^2(1-q)^2 \{I(q) - I(1-q)\} \begin{cases} > 0, & q > 1/2, \\ = 0, & q = 1/2, \\ < 0, & q < 1/2. \end{cases}$$

Thus the best sensor is plus above $1/2$ and minus below it.

The first two choices are symmetric from preparation $(1/2, 1/2)$, so fix the first sensor as plus. Its report density and next predicted probability are

$$r_0(y) = \frac{1 + \lambda y/2}{2}, \quad q(y) = \alpha + (1 - 2\alpha) \frac{1 + \lambda y}{2 + \lambda y}.$$

In particular $q(y) - 1/2$ has the sign of y . The two quantities to compare are

$$(23) \quad R_{\text{fixed}}^{(2)} = \min_{a \in \{+, -\}} \int_{-1}^1 b_a(q(y)) r_0(y) dy,$$

$$(24) \quad R_{\text{feedback}}^{(2)} = \int_{-1}^1 \min_{a \in \{+, -\}} b_a(q(y)) r_0(y) dy.$$

Their difference is the smaller of the strictly positive integrals of $b_+ - b_-$ on $(-1, 0)$ and $b_- - b_+$ on $(0, 1)$, weighted by r_0 . This also compares with sequences starting with minus, by complementing

the hidden state and exchanging sensor names. For an explicit bound, the $k = 1$ term of (21) and positivity of all later differences give

$$R_{\text{fixed}}^{(2)} - R_{\text{feedback}}^{(2)} \geq \frac{\alpha^2(1-\alpha)^2(1-2\alpha)\lambda^5(1-\lambda/2)}{20(2+\lambda)} > 0.$$

Indeed on each half-interval $|y| \in [1/2, 1]$, one has $q(1-q) \geq \alpha(1-\alpha)$, $|2q-1| \geq (1-2\alpha)\lambda/[2(2+\lambda)]$, and $r_0 \geq (1-\lambda/2)/2$. This proves the strict gain without a numerical experiment. Positive transition entries and positive likelihoods leave both hidden states with positive posterior mass after every report. The final memory assertion follows from Theorem 2.2 with $d = 1$. $\square$

This is controlled hidden-state sensing with continuous, overlapping emissions. The all-horizon theorem optimizes every declared sensor choice by (7); a closed formula for that optimal policy is not assumed. The two-call calculation proves that the class actually contains observation-dependent optimal actions. It is not an assertion that a threshold of this form remains optimal at every longer horizon. The model is a calibrated finite-state sensor family, not an unknown-kernel or arbitrary nonlinear-filter theorem.

4. SINGULAR ACQUIRED LAWS AND CHANGING STRATA

The density assumption may be replaced by an explicitly stratified acquired law. Allow normalized nonnegative report kernels, including perfectly informative marked reports. Bayes updates are defined on the events of positive report probability; arbitrary definitions on null events have no effect. Let $\mathcal{S}$ be a finite collection of distinct simplex faces, including possible vertices. For $s \in \mathcal{S}$ write d_s for its affine dimension. Fix a compact convex K_s in the relative interior of that face, with a fixed relative-interior margin. Different relative interiors are disjoint, so the stratum is a Borel function of the posterior. It is not supplied as an uncharged retained register.

Theorem 4.1 (Stratified posterior pooling). *Suppose uniformly in p, a that every actual one-step posterior law decomposes as*

$$(25) \quad \Lambda_{t-1}(p, a) = \sum_{s \in \mathcal{S}} \Lambda_{t,s}(p, a), \quad \text{supp } \Lambda_{t,s}(p, a) \subset K_s.$$

For $d_s > 0$ suppose $\Lambda_{t,s}(p, a)$ has relative-coordinate density at most $\rho_{t,s}$. These are subprobability laws: $\rho_{t,s}$ includes their actual mass. A zero-dimensional component is supported on its one vertex.

Choose sufficiently large integers $L_{t,s}$ for each positive-dimensional stratum and one label for each vertex. There is a causal observer of risk

$$(26) \quad R - B_n^* \leq C \sum_{t=1}^n \sum_{s: d_s > 0} \rho_{t,s} L_{t,s}^{-2/d_s}$$

with at most

$$(27) \quad 1 + \sum_{t=1}^n \left(\#\{s : d_s = 0\} + \sum_{s: d_s > 0} L_{t,s} \right)$$

total states. Its excess equals the stratumwise version of (11). The constants depend only on the finite relative geometries and the terminal loss.

If a terminal stratum s_ of dimension $r > 0$ has mass at least $w > 0$ under every conditional last-step law and density at most ρ_* , then every checkpoint or autonomous M -state procedure has*

$$(28) \quad R - B_n^* \geq c_r w^{1+2/r} \rho_*^{-2/r} M^{-2/r}.$$

In particular, if all other stratum dimensions are at most r , checkpoint and autonomous memory exponents match at $2/r$ for each fixed horizon. No uniformity is claimed as w vanishes or a relative-interior margin degenerates.

Proof. Keep distinct labels for different faces, and partition each positive-dimensional K_s by a grid in its affine coordinates. Pool all incoming laws in each of these cells. Their conditional barycenters stay in the same compact convex face subset. Lemma 2.4 therefore gives exact retained posteriors just as before. Restrict the concave V_t to that face; it remains uniformly Lipschitz in relative coordinates. Lemma 2.3 applies to each actual submeasure without renormalization. Vertex components have zero Jensen defect because all their posterior values are the same vertex. Sum the defects to obtain (26). The labels (t, s, j) and the initial state give exactly the count (27), so the phase and stratum are not free.

Under any full-history policy the terminal s_* -submeasure still has mass at least w and density at most ρ_* . Project any M centers onto its affine span. Their radius- u balls cover mass at most $\rho_* v_r M u^r$. Choosing $u = (w/(2\rho_* v_r M))^{1/r}$ leaves at least $w/2$ mass uncovered. Projection and conditional squared-loss decomposition give (28). An equal constant-fraction allocation of the available labels to the finitely many stages and positive-dimensional strata gives exponent $2/r$ in the upper, since $L^{-2/d_s} \leq L^{-2/r}$ for $d_s \leq r$. The fixed-horizon checkpoint upper follows by applying the same stratified terminal grid to an optimal full-history policy. This proves both matching assertions without deriving an online lower from terminal geometry when an intermediate dimension is larger than r . $\square$

Corollary 4.2 (Intermittently validated controlled sensing). *At call t of the two-state sensor in Theorem 3.1, independently choose a validated report with calibrated probability $\theta_t \in [0, 1]$. A validated report physically reveals the new hidden state; otherwise the selected overlapping sensor emits its continuous report. The validation flag is part of this single paid report. Then the acquired posterior law is the sum of vertex atoms of total mass θ_t and a continuous component of mass exactly $1 - \theta_t$ and density at most $(1 - \theta_t)\rho$.*

A controller using L_t continuous cells and two vertex labels per stage has $1 + \sum_t (L_t + 2)$ total states and

$$(29) \quad R - B_n^* \leq C\rho \sum_{t=1}^n (1 - \theta_t) L_t^{-2}.$$

If $\theta_n < 1$, both checkpoint and autonomous excesses are bounded below by

$$(30) \quad c(1 - \theta_n)\rho^{-2}M^{-2}.$$

For fixed n with θ_n bounded away from one, the memory exponent is two. No lower floor survives when the terminal call validates the state with probability one.

Proof. On a validated report the posterior is exactly 0 or 1, not a formal limiting state. On a nonvalidated report its law is that of Theorem 3.1. The independent validation probability gives the stated actual masses. A positive physical transition carries a vertex posterior to a predicted probability in $[\alpha, 1 - \alpha]$, so the next noisy report can again acquire a continuous posterior. Thus the experiment repeatedly moves between zero- and one-dimensional acquired strata. Apply Theorem 4.1 with two vertices and the compact noisy interval. In (28), substitute $w = 1 - \theta_n$ and $\rho_* = (1 - \theta_n)\rho$ to obtain (30). $\square$

For two calls, if both validation probabilities are less than one, the strict feedback gain from Theorem 3.1 remains at least $(1 - \theta_1)(1 - \theta_2)$ times its displayed bound. The nonvalidated first-report event already gives that gain; validated histories give a nonnegative additional benefit to the optimized feedback policy. This singular extension does not leave changing rank as an interface assertion, and rare-stratum mass is visible in both bounds.

5. ONE-TASK COMPOSITION AND FINITE IMPLEMENTATION

5.1. Typed causal comparison. A resource-certified morphism $\mathfrak{E} \rightsquigarrow \mathfrak{F}$ consists of a parameter-independent causal simulator, its internal state, a lift of each strategy in a *specified* target strategy class, report transformations, clock and action-compatibility rules, and the scored-task readout. The simulated scored law, including public actions, reports, costs and the audit, must have total-variation distance at most ε from the target law, uniformly over the declared parameters and target strategies. It is not enough to match single report marginals. A task posterior smaller than the full predictive quotient does not, by itself, define this simulator.

Proposition 5.1 (Common-task risk and product resources). *Suppose $\mathfrak{E} \rightsquigarrow \mathfrak{F}$ has the preceding certificate for a loss in $[0, L_{\text{loss}}]$. A target controller with M total states, N calls, scratch W and programme description P has a source implementation with at most MSQ states, where S is the simulator-state count and Q counts any additional genuinely retained clock/interface states. Its risk increases by at most $L_{\text{loss}}\varepsilon$. Calls and physical duration follow the certified functions $C_{\text{call}}(N)$ and $C_{\text{time}}(N)$; scratch is at most $W + W_{\text{sim}} + W_{\text{buffer}}$, and programme descriptions add with the composition instructions.*

Independent stagewise kernel errors bounded by ζ_t are not required: uniform conditional joint-kernel TV bounds alone add at most $\sum_t \zeta_t$ to the scored-law defect. Serial morphism defects add, with the respective strategy lifts composed. A reverse risk inequality requires a reverse certificate covering every source strategy being lower bounded, and uses its own resource map. All these are inequalities for the same absolute scored risk, not differences of separate Bayes baselines.

Proof. Store the target, simulator and additional interface states as one tuple; causality and action compatibility make its update legal. Couple each next simulated joint operation maximally until the first mismatch. A union bound gives the accumulated stagewise defect even for adaptive strategies, since the bound is conditional and uniform. Composition and data processing give the serial assertion. For a scored loss in $[0, L_{\text{loss}}]$, the change in its expectation is at most L_{loss} times TV. Apply this to each certified target strategy before taking its infimum. For a lower bound reverse the construction and verify that the reversed lift covers the entire class whose infimum is used. No optimal-control comparison outside those classes follows. $\square$

5.2. Numerical evaluation without a hidden posterior register.

Proposition 5.2 (Certified digital posterior pooling). *In Theorem 2.2, allow the programmed action at stage t to be within γ_t of its Bellman minimum at that label's exact conditional posterior. Recompute the pooling barycenters under this programmed controller's actual law. Its ideal risk upper gains only $\sum_t \gamma_t$.*

Let h_t be the side length of its grid. Suppose a finite evaluation of its fixed programme satisfies the following certificates: outside a failure event of probability at most ξ_t , its computed posterior coordinates differ from the ideal coordinates by at most $\beta_t < h_t/4$; its terminal readout lies in the simplex and differs from the ideal barycenter in task norm by at most u ; and the physical calibrated joint kernel differs from the reference kernel by at most ζ_t in uniform conditional TV. Then

$$(31) \quad R_{\text{digital}} \leq B_n^* + C \sum_t L_t^{-2/d} + \sum_t \gamma_t + u^2 + 2 \sum_t \left(\zeta_t + \xi_t + C_d \rho \frac{\beta_t}{h_t} \right).$$

An additional certified causal simulation defect adds 2ε and multiplies the state count as in Proposition 5.1. The grid label count is unchanged; all finite evaluation workspace is erased at the next cut.

Proof. A γ_t -optimal action changes the Bellman equality by a nonnegative term at most γ_t . The exact aggregation argument and the Jensen estimate are otherwise unchanged. Couple the ideal

and digital controllers until their first label mismatch. While their labels agree they issue the same recorded action. A coordinate error of at most β_t changes a grid label only when the ideal posterior is within β_t of a grid hyperplane. In the unit coordinate cube there are at most C_d/h_t such hyperplanes, so their slabs have actual mass at most $C_d\rho\beta_t/h_t$. Bound the first mismatch by the sum of these probabilities and the certified failure/kernel-error probabilities. This uses the ideal unconditioned acquired law; restricting to histories before the first mismatch can only decrease those event probabilities. No conditional density after that restriction is assumed.

With ideal transitions and a rounded terminal output, projection at the terminal label adds exactly its squared readout error, at most u^2 . Couple to this rounded-output reference controller. Both forecasts lie in the simplex, so their scored loss lies in $[0, 2]$; the mismatch probability therefore costs at most twice its bound. This proves (31) and the final morphism assertion. $\square$

The different precisions in this proposition are intentional. Bellman comparison error is in risk units; posterior evaluation error is amplified near cell boundaries by h_t^{-1} ; terminal-output error is squared. A permitted numerical tolerance is not automatically an unavoidable risk floor.

For the rational sensor kernels of Theorem 3.1, denominators in (19) are bounded below by $1 - \lambda$. A stored coefficient error u_{coef} , report-coordinate error u_{report} and arithmetic error u_{arith} give $\beta_t \leq C(u_{\text{coef}} + u_{\text{report}} + u_{\text{arith}})$. Thus, for fixed n, d and a common mesh h , these three errors may be chosen $O(h^3)$, Bellman comparison errors $O(h^2)$ and output error $O(h)$ to preserve the h^2 risk order. The comparison errors and barycenter approximations must be certified in preprocessing; no decidability claim is made for arbitrary Borel constants.

For $S = 1 + \sum_t L_t$, a finite programme records one action and one posterior coefficient vector per state, stage/grid data, and terminal outputs. With p_{coef} coefficient bits and p_{out} output bits per coordinate, a coarse description bound is

$$O\left(S\{(d+1)p_{\text{coef}} + \log(|\mathcal{A}| + 1) + \log(n+1) + \log(\bar{L} + 1)\} + (d+1)L_n p_{\text{out}} + P_K\right),$$

where P_K is the known kernel-evaluation programme. Here $\bar{L} = \max_t L_t$ is the largest per-stage label allocation. Scratch and kernel-evaluation costs are charged separately; a sequential table scan avoids any assumption of unit-cost random access. For fixed d , ordinary fixed-precision arithmetic uses polynomial scratch and bit time in the coefficient, report and arithmetic precisions per scan, and at most n such scans of S records. The interpreter resets its position before each persistent cut. There is no stored real-valued filter beyond the label. Planning may be exponential or require large certified integrations, even though the deployed programme is finite.

5.3. An attained joint profile.

Corollary 5.3 (Calibration and early-channel deficiency in one task). *Adjoin to the controlled sensor experiment an unknown bit $b \in \{0, 1\}$ and a calibration sign $\sigma \in \{-1, 1\}$. Both are independent of its prepared hidden-state randomness. The known amplitude is $0 \leq \delta \leq 1$. Before sensing, an early channel reports b with probability $1 - 2\varepsilon$ and an erasure symbol otherwise, where $0 \leq \varepsilon \leq 1/2$. The common terminal audit is*

$$(e_{X_n}, b, \delta\sigma),$$

*with direct-sum squared loss. The sign is never reported before this audit. The exact-bit target requires its first virtual report to be completed **before all sensing calls and before the terminal audit**. Under this order the causal deficiency of the early channel relative to the exact-bit target is exactly ε .*

Let $\mathcal{R}$ be minimax over (b, σ) , with the hidden preparation fixed. For all sufficiently large total-state budgets,

$$(32) \quad \begin{aligned} cM^{-2/d} + \delta^2 + \varepsilon/2 &\leq \mathcal{R}(n+3, M, \delta, \varepsilon) - B_n^* \\ &\leq Cn \left\lfloor \frac{M-4}{3n} \right\rfloor^{-2/d} + \delta^2 + \varepsilon/2. \end{aligned}$$

There are n sensing calls, one preparation, one early-channel call and one joint audit; their physical durations remain separately charged. In particular, for each fixed n the matched excess profile is $M^{-2/d} + \delta^2 + \varepsilon$. This is not a sharp joint large- n theorem.

Proof. Average the lower bound over independent fair b and σ . Conditional on every channel status, all sensing strategies remain covered by Theorem 2.2; unrelated side information cannot improve B_n^* , and the terminal forecast still has at most M centers. The hidden-coordinate risk is therefore at least $B_n^* + cM^{-2/d}$. The sign is independent of every pre-audit observation, so its best squared risk is δ^2 . On erasure the bit remains fair even after the sensing history, giving bit risk $(2\varepsilon)/4 = \varepsilon/2$. The loss is a direct sum, so these lower bounds add under the same product prior; minimax risk dominates its Bayes risk.

For the upper, retain one of three channel statuses and a pooling controller with $1 + nL$ states, where $L = \lfloor (M-4)/(3n) \rfloor$. The initial channel state plus this product uses at most $1 + 3(1 + nL) = 4 + 3nL \leq M$ states. Forecast the known bit on a nonerasure, $1/2$ on erasure and zero for the sign. Theorem 2.2 gives the upper. Retaining the status throughout is a safe count, not a claim that no states can be merged.

A simulator outputs the received bit or guesses a fair bit on erasure, completing that report in the required early phase. Coupling shows full scored-law defect at most ε even when later target actions depend on the virtual bit. For the reverse bound, the two early source laws have TV $1 - 2\varepsilon$, whereas the two exact target-bit laws have TV one. Data processing and the triangle inequality force every parameter-independent simulator's worst-parameter defect to be at least ε . The early-completion rule is essential: the terminal audit is not available when producing this report. $\square$

The calibration and erasure coordinates are inherited elementary constructions; the new assertion is their composition with the overlapping-noise adaptive completion and its complete state count. General calibration errors in a physical kernel obey Proposition 5.2, not automatically the particular δ^2 law of this inaccessible-sign experiment.

6. RELATION TO EARLIER THEORIES AND THE RETAINED MECHANISMS

The projection identity (3), belief-state reduction, concavity of finite-horizon Bayes risk, conditional expectation and positive-kernel comparison are standard tools. The contribution here is the exact pooling identity (11) combined with integrated curvature, actual posterior-density converses, adaptive action selection and a complete autonomous state count. Theorem 2.2 is not a claim that finite-memory control, belief quantization or active sensing is new.

6.1. Finite-memory control and quantization. Kara and Yüksel [5] approximate a partially observed control problem by a finite window of actions and reports. Under filter stability they establish policy-error bounds in terms of window length; their finite-action, finite-observation construction admits general hidden state spaces. Our reports are continuous and a retained label is not a stored observation window. Lemma 2.4 computes its true conditional hidden distribution under the finite observer's own law. The hypotheses are different: posterior smoothing and an interior carrier replace the stability hypothesis, while the conclusion matches a terminal squared-risk memory exponent and counts the internal phase. Neither theorem contains the other.

Approximate information states [16] control planning loss through reward and transition approximation. Belief covering [19] makes geometric complexity relevant to POMDP planning, and filter quantization [12] provides recursive finite approximations of conditional distributions. Here a pointwise approximation error does not suffice: it would give a first-order bound at a nonsmooth policy switch. The zero conditional first moment in (15), together with integration of the curvature measure, gives a second-order bound even at such switches. Moreover, the representatives are not required to approximate the full-history posterior pathwise. Static quantization [3] supplies the classical terminal problem; it does not itself give this causal realization.

6.2. Observation-adaptive sensing. Adaptive submodularity [2] gives approximation guarantees for adaptive stochastic utility optimization and coverage. It concerns the value or test cost of a policy, not the number of distinct states in a persistent controller. No submodularity of the present squared audit risk is asserted. For coverage comparisons one must distinguish the corrected author version’s additional assumptions and squared-logarithmic guarantee from the original, subsequently corrected theorem.

Active sequential hypothesis testing [9] balances acquisition and decision costs through sequential policies and information bounds. Controlled sensing with Markovian observations [10] permits causal sensing, nonuniform costs and asymptotic testing guarantees. Those tasks optimize stopping and identification rather than a fixed-horizon categorical forecast under a hard internal-state budget. Theorem 2.2 instead fixes the physical horizon, compares against the optimal full-history sensor policy, and supplies a memory converse in the same loss. It does not replace their sequential or asymptotic optimality assertions.

Noisy optimal decision trees [4] optimize test cost for identifying a hypothesis, including persistent noisy outcomes. Their objective and noise model differ from squared hidden-state prediction. A policy tree and a finite persistent observer are not synonymous: every internal node is part of the autonomous state count unless a separate implementation merges it. Supplement T’s separated-refinement proof explicitly pays the entire tree; the present observer merges incoming edges by conditional barycenters and pays the entire phase-labelled state set. Existence of that programme is not an efficient decision-tree synthesis theorem.

Active feature acquisition also changes the next observation with the already acquired values. The acquisition-conditioned oracle [17] constructs nongreedy acquisition policies for predictive tasks. The time-varying evaluation theory of von Kleist et al. [6] addresses distribution shifts induced by changing acquisition policies, using explicit causal assumptions. We likewise distinguish the actual law under each policy, but assume known calibrated kernels and prove a finite-state geometric risk law; we do not estimate its policy value from retrospectively acquired data. In particular (11) uses the law of the implemented controller, not data distributed according to another exploration policy.

6.3. Prediction, comparison and causal information. Predictive-state representations [8] and causal-state constructions [15] motivate the future-test equivalence relation. Appendix A states the additional measurable realization conditions used here; a formal quotient is not silently assumed standard Borel. Blackwell comparison and statistical deficiency [1, 7] supply the classical decision-theoretic basis for Proposition 5.1. The additional causal assertion requires the simulator’s actual internal memory, strategy lift, clock and joint audit law.

The retained continuation-cut bounds in Supplement T are finite functional coding problems with decoder side information. Their relationship with Wyner–Ziv and coding for computing [18, 11] is retained there. They expose intermediate information that terminal quantization can miss. The present pooling identity is instead an upper implementation theorem for adaptive controlled hidden-state experiments, with a direct actual-mass lower. It is not a proof that arbitrary continuation widths always generate a recursion.

6.4. What has and has not been unified. The separated-refinement completion theorem in Supplement T requires positive branch mass, global covering and a response-support gap; it proves a horizon-uniform two-budget comparison. Theorem 2.2 removes that support gap on a positive noisy sensor class and gives a full-history adaptive comparison, but retains the explicit horizon factor in (12). The serial continuation and sparse-filter theorems in the integral supplements remain independent sufficient mechanisms, with all their proofs. Their fixed-controller or fixed-acquisition assertions have not been promoted to unrestricted optimal control.

A unifying decision principle is now explicit: forgetting replaces a posterior by a conditional barycenter, and its exact loss is a Jensen defect of the optimal executable continuation value. Separated geometry controls a related loss by actual-mass tree frontiers; smooth acquired laws control it by integrated curvature. This principle does not yet imply a near-necessary characterization, horizon-uniform matching under overlapping noise, unknown-kernel learning, or optimal programme, workspace and planning complexity. Those conclusions require further proofs, not reinterpretation of the present inequalities.

APPENDIX A. MEASURABLE PREDICTIVE REALIZATION

We record a sufficient realization result so that the predictive quotient in Section 1 is an actual measurable object. Its topological hypotheses are explicit; they are not inferred from the phrase “standard Borel”. The more general instrument construction and its hypotheses are retained in Supplements S and T.

Proposition A.1 (Compact evaluation realization). *At each fixed phase let the history space be a countable union of Borel compact metrizable pieces $H = \bigcup_m H_m$. Suppose a countable determining family of bounded executable tests and legal-interface coordinates has continuous normalized expectations $e_k : H_m \rightarrow [-1, 1]$ on every piece. Assume equality of all these coordinates is equivalent to equality of the legal interface and expectations of every executable continuation test. Put*

$$e(h) = (e_k(h))_{k \geq 1}, \quad S = e(H) \subset [-1, 1]^{\mathbb{N}}.$$

Then S is standard Borel, e admits a Borel section, and its fibers are exactly predictive equivalence classes. Every Borel history statistic b from which e is Borel recoverable factors onto S ; conversely every Borel statistic constant on the fibers is a Borel function of e .

Suppose additionally that the report kernel and the updated evaluation coordinates after each legal action are jointly Borel in history, action and report and are constant on each predictive fiber (for a fixed jointly defined version). Then they descend to jointly Borel kernels and updates on S . The latter assumption holds for the finite-state Bayes realization (5) whenever the future tests separate the relevant beliefs and legal phase is retained.

Proof. Each image $e(H_m)$ is compact, so S is an F_σ subset of the compact countable product. A continuous surjection from a compact metric space onto its image has a Borel section: cover the domain at level r by finitely many closed sets of diameter at most 2^{-r} , refine the previous chosen closed set, and select the first set whose image contains the desired point. Images of these compact sets are closed. The selected nested nonempty compact sets have diameters tending to zero, and their unique limit is a Borel selection in the fiber. For S , first select the least piece whose compact image contains the point, then use its section. This supplies a Borel map $s : S \rightarrow H$ with $e \circ s = \text{id}_S$.

The determining-family assumption identifies the fibers exactly, including action availability. If $e = g \circ b$, the statistic b maps onto S through g ; this is the predictive minimality statement. If v is Borel and constant on the fibers, $v = (v \circ s) \circ e$, giving Borel factorization. The report kernel and update descend by evaluation at $s(z)$. Fiber constancy makes the definition independent of the

section; joint measurability follows by composition. No assertion is made about arbitrary null-set versions of regular conditional distributions without the stipulated compatible version. $\square$

For the sensors of Theorem 3.1, fixed-phase histories form finitely many compact report cubes indexed by action words. Positivity makes Bayes evaluation continuous on each. If an invertible legal transition is available, next-report means recover the incoming belief as proved there. Terminal categorical audit tests separate terminal beliefs. Equality of beliefs gives equality of every controlled future law by iteration of the physical kernels, so the quotient is the actual acquired belief image, with phase appended. For a parameter family these statements apply to the declared prepared law; a Bayesian quotient for one prior is not automatically a sufficient statistic valid simultaneously for every parameter.

A smaller state used only for a scored task need not satisfy the determining-family premise for the complete experiment. In particular a continuation chart, an observable rank, or a scalar forecast does not by itself yield a simulator of all future tests. Simulator certification in Proposition 5.1 therefore remains a separate requirement. Nothing here extends a finite-dimensional identity to an unbounded operator closure or converts total-variation comparison into large-deviation equivalence.

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